Exercise 1a

The full hamiltonian for the system is given by

$$\mathcal{H} = \mathcal{H}_{\text{atom}} + \mathcal{H}_{\text{int}} + \mathcal{H}_{\text{field}}$$

Let $|i\rangle\langle j| = |i\rangle \otimes \langle j_k|$ be a basis for the system with $|i\rangle$ being eigenstates of the atom and $|j_k\rangle$ representing a field with j photons in mode k. Setting $|1\rangle$ to be zero energy, the atomic hamiltonian is given by

$$\mathcal{H}_{\text{atom}} = \hbar\omega_{12}|2\rangle\langle 2| + \hbar\omega_{13}|3\rangle\langle 3|$$

The radiation field has hamiltonian

$$\mathcal{H}_{\text{field}} = \sum_{k,\lambda} \hbar\nu_{k,\lambda}\left(a^\dagger_{k\lambda}a_{k\lambda} + \frac{1}{2}\right)$$

and the interaction hamiltonian is given by

$$\mathcal{H}_{\text{int}} = -\hat{d}\cdot\hat{E} = -\sum_{\alpha,\alpha'}|\alpha\rangle\langle\alpha|\hat{d}|\alpha'\rangle\langle\alpha'| \cdot \sum_{k,\lambda}\sqrt{\frac{\hbar\nu_k}{2\varepsilon_0 V}}\left(\hat{a}_{k\lambda}\hat{\varepsilon}_{k\lambda}e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger_{k\lambda}\varepsilon^*_{k\lambda}e^{-i\vec{k}\cdot\vec{r}}\right)$$

By assumption, $\langle 2|\hat{d}|1\rangle = \langle 3|\hat{d}|1\rangle = \mu\hat{z}$. Letting $g_k = \frac{\mu}{\hbar}\sqrt{\hbar\nu_k/2\varepsilon_0 V}(\hat{\varepsilon}_k\cdot\hat{z})e^{i\vec{k}\cdot\vec{r}}$, we get

$$\mathcal{H}_{\text{int}} = -\hbar\sum_{k,\lambda} g_k\left(|2\rangle\langle 1| + |3\rangle\langle 1|\right)\hat{a}_{k\lambda} + g^*_k\left(|1\rangle\langle 2| + |1\rangle\langle 3|\right)\hat{a}^\dagger_{k\lambda}$$

To study spontaneous emission, we consider states $|2,0\rangle, |3,0\rangle$ (atom in excited state, no photons) and $|1,1_k\rangle$ (atom in ground state after emitting photon in mode k). We evaluate the matrix elements as

$$\begin{aligned}
\langle 2,0|\mathcal{H}|2,0\rangle &= \hbar\omega_{12} & \langle 3,0|\mathcal{H}|3,0\rangle &= \hbar\omega_{13}\\
\langle 1,1_k|\mathcal{H}|1,1_k\rangle &= \hbar\nu_k & \langle 2,0|\mathcal{H}|1,1_k\rangle &= -\hbar g_k\\
\langle 3,0|\mathcal{H}|1,1_k\rangle &= -\hbar g_k
\end{aligned}$$

where we ignored the $\hbar\nu_k/2$ term in the field hamiltonian since it adds a constant to the total energy, irrelevant to dynamics. Thus, we get

$$\mathcal{H} = \underbrace{\hbar\omega_{12}|2\rangle\langle 2| + \hbar\omega_{13}|3\rangle\langle 3| + \sum_k \hbar\nu_k|k\rangle\langle k|}_{:=\mathcal{H}_0} - \sum_k(\hbar g_k|2\rangle\langle k| + \hbar g_k|3\rangle\langle k| + \text{h.c.})$$

where we dropped the sum over λ by absorbing it implicitly into the sum over k. Let $\mathcal{H}_1$ be the interaction part of the hamiltonian, i.e $\mathcal{H}_1 = \mathcal{H} - \mathcal{H}_0$. We move to the

interaction picture w.r.t $\mathcal{H}_0$ by defining $|\psi_I\rangle = \hat{U}^{-1}|\psi\rangle$ where $\hat{U} = e^{-i\mathcal{H}_0 t/\hbar}$. Then,

$$\begin{aligned} i\hbar\frac{d|\psi_I\rangle}{dt} &= i\hbar\frac{d\hat{U}^{-1}}{dt}|\psi\rangle + i\hbar\hat{U}^{-1}\frac{d|\psi\rangle}{dt} \\ &= -\mathcal{H}_0\hat{U}^{-1}|\psi\rangle + \hat{U}^{-1}(\mathcal{H}_0+\mathcal{H}_1)|\psi\rangle \\ &= \hat{U}^{-1}\mathcal{H}_1|\psi\rangle \\ &= \hat{U}^{-1}\mathcal{H}_1\hat{U}|\psi_I\rangle \end{aligned}$$

where we used the fact that $\hat{U}^{-1}$ commutes with $\mathcal{H}_0$ in the last step. Thus, our effective hamiltonian in the interaction picture is $\mathcal{H}_{\text{eff}} = \hat{U}^{-1}\mathcal{H}_1\hat{U}$. We note that since $\mathcal{H}_0$ is diagonal, we have

$$\hat{U}^{-1} = e^{i\mathcal{H}_0 t/\hbar} = e^{i\omega_{12}t}|2\rangle\langle 2| + e^{i\omega_{13}t}|3\rangle\langle 3| + \sum_k e^{i\nu_k t}|k\rangle\langle k|$$

Hence, we get

$$\mathcal{H}_{\text{eff}} = -\hbar\sum_k g_k|2\rangle\langle k|e^{-i(\nu_k-\omega_{12})t} + g_k|3\rangle\langle k|e^{-i(\nu_k-\omega_{13})t} + \text{h.c.}$$

Exercise 1b

Let $|\psi_I(t)\rangle = c_2(t)|2\rangle + c_3(t)|3\rangle + \sum_k c_k(t)|k\rangle$. We want to remove the explicit time dependence in $\mathcal{H}_{\text{eff}}$. First, we bring up the energy of the $|k\rangle$ by making the unitary transformation $|\overline{\psi}\rangle = U_1|\psi\rangle$, with $U_1 = e^{-i\sum_k(\nu_k-\omega_{13})t|k\rangle\langle k|} = \sum_k e^{-i(\nu_k-\omega_{13})t}|k\rangle\langle k|$. The last step holds since $|k\rangle$ is a basis element so $\sum_k|k\rangle\langle k|$ is diagonal. We see that

$$i\hbar\frac{d|\overline{\psi}\rangle}{dt} = i\hbar\left(\frac{dU_1}{dt}|\psi\rangle + U_1\frac{d|\psi\rangle}{dt}\right) = \underbrace{(i\hbar\frac{dU_1}{dt}U_1^\dagger + U_1\mathcal{H}_{\text{eff}}U_1^\dagger)}_{:=\overline{\mathcal{H}}_{\text{eff}}}|\overline{\psi}\rangle$$

We have

$$i\hbar\frac{dU_1}{dt}U_1^\dagger = i\hbar\left(\frac{d}{dt}e^{-i\sum_k(\nu_k-\omega_{13})t|k\rangle\langle k|}\right)U_1 = \hbar\sum_k(\nu_k-\omega_{13})|k\rangle\langle k|$$

and

$$U_1\mathcal{H}_{\text{eff}}U_1^\dagger = -\hbar\sum_k g_k|2\rangle\langle k|e^{-i\omega_{23}t} + g_k|3\rangle\langle k| + \text{h.c.}$$

where $\omega_{23} = \omega_{13} - \omega_{12}$. Hence,

$$\overline{\mathcal{H}}_{\text{eff}} = \hbar\sum_k(\nu_k-\omega_{13})|k\rangle\langle k| - \hbar\sum_k g_k|2\rangle\langle k|e^{-i\omega_{23}t} + g_k|3\rangle\langle k| + \text{h.c.}$$

To get rid of the $e^{-i\omega_{23}t}$ dependence, we bring down state $|2\rangle$ by ω_{23} with the unitary transform $|\widetilde{\psi}\rangle = U_2|\overline{\psi}\rangle$ with $U_2 = e^{i\omega_{23}t|2\rangle\langle 2|} = e^{i\omega_{23}t}|2\rangle\langle 2|$ since $|2\rangle$ is a basis element. Then, similar as before, we find

$$\widetilde{\mathcal{H}}_{\text{eff}} = -\hbar\omega_{23}|2\rangle\langle 2| + \hbar\sum_k(\nu_k - \omega_{13})|k\rangle\langle k| - \hbar\sum_k g_k|2\rangle\langle k| + g_k|3\rangle\langle k| + \text{h.c.}$$

Now, we look at the schrodinger equation, $i\hbar\frac{d|\widetilde{\psi}\rangle}{dt} = \widetilde{\mathcal{H}}_{\text{eff}}$, where $|\widetilde{\psi}\rangle = \tilde{c}_2(t)|2\rangle + \tilde{c}_3(t)|3\rangle + \sum_k \tilde{c}_k(t)|k\rangle$, to get

$$\begin{aligned}
\dot{\tilde{c}}_2 &= i\omega_{23}\tilde{c}_2 + i\sum_k g_k\tilde{c}_k \\
\dot{\tilde{c}}_3 &= i\sum_k g_k\tilde{c}_k \\
\dot{\tilde{c}}_k &= i\underbrace{(\omega_{13} - \nu_k)}_{:=\delta_k}\tilde{c}_k + ig_k^*\tilde{c}_2 + ig_k^*\tilde{c}_3
\end{aligned}$$

We integrate $\dot{\tilde{c}}_k$ using the integrating factor $e^{-i\delta_k t}$ to get

$$c_k(t) = ig_k^*\int_0^t e^{-i\delta_k(t'-t)}(\tilde{c}_2(t') + \tilde{c}_3(t'))dt'$$

where we imposed the intial condition $\tilde{c}_k(0) = 0$. We assume $\tilde{c}_2$ and $\tilde{c}_3$ vary slowly so that

$$c_k(t) = ig_k^*(\tilde{c}_2(t) + \tilde{c}_3(t))\int_0^t e^{-i\delta_k(t'-t)}dt' = ig_k^*(\tilde{c}_2(t) + \tilde{c}_3(t))(\pi\delta(\delta_k) + i\mathcal{P}\frac{1}{\delta_k})$$

Plugging this into $\tilde{c}_2$ gives us

$$\dot{\tilde{c}}_2 = i\omega_{23}\tilde{c}_2 - (\tilde{c}_2 + \tilde{c}_3)\sum_k |g_k|^2(\pi\delta(\delta_k) + i\mathcal{P}\frac{1}{\delta_k})$$

The $i\mathcal{P}(1/\delta_k)$ term leads to energy shifts (lamb shifts in this context) which we can ignore. Invoking results form the notes, we evaluate

$$\sum_k |g_k|^2\pi\delta(\delta_k) = \frac{\mu^2\omega_{13}^2}{3\pi\varepsilon_0\hbar c^3} \approx \frac{\mu^2\omega_{12}^2}{3\pi\varepsilon_0\hbar c^3} := \frac{\Gamma}{2}$$

where was assume $\omega_1 3 \approx \omega_1 2$. All together, we get

$$\dot{\tilde{c}}_2 = i\omega_{23}\tilde{c}_2 - (\tilde{c}_2 + \tilde{c}_3)\Gamma/2 = (i\omega_{23} - \Gamma/2)\tilde{c}_2 - (\Gamma/2)\tilde{c}_3$$

For $\dot{\tilde{c}}_3$, we similarly get

$$\begin{aligned}
\dot{\tilde{c}}_3 &= -(\tilde{c}_2 + \tilde{c}_3)\sum_k |g_k|^2(\pi\delta(\delta_k) + i\mathcal{P}\frac{1}{\delta_k}) \\
&= -(\Gamma/2)\tilde{c}_3 - (\Gamma/2)\tilde{c}_2
\end{aligned}$$

So, we get

$$\dot{\tilde{c}}_2 = (i\omega_{23} - \Gamma/2)\tilde{c}_2 - (\Gamma/2)\tilde{c}_3 \tag{1a}$$

$$\dot{\tilde{c}}_3 = -(\Gamma/2)\tilde{c}_3 - (\Gamma/2)\tilde{c}_2 \tag{1b}$$

Exercise 1c

We note that unitary transformation preserve inner products as if $|\tilde{a}_i\rangle = U|a_i\rangle$ then $\langle\tilde{a}_i|\tilde{a}_j\rangle = \langle a_i|U^\dagger U|a_j\rangle = \langle a_i|a_j\rangle$. So $\tilde{c}_2 = \langle\tilde{2}|\tilde{\psi}\rangle = \langle 2|\psi\rangle = c_2$.
We have a planar system for $\tilde{c}_2, \tilde{c}_3$, with a coefficient matrix

$$A = \begin{pmatrix} i\omega_{23} - \Gamma/2 & -\Gamma/2 \\ -\Gamma/2 & -\Gamma/2 \end{pmatrix}$$

The time evolution of our amplitues is then

$$\begin{pmatrix} \tilde{c}_2(t) \\ \tilde{c}_3(t) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{At}$$

with the initial conditions $\tilde{c}_2(0) = 1$ and $\tilde{c}_3(0) = 0$. We solve this using mathetmatica (see accompanying notebook), to get

$$\Gamma \geq \omega, \qquad |c_2(t)|^2 = \frac{e^{-t\Gamma}((1+\cosh(\Omega t))\Gamma^2 - 2\omega^2)}{2\Omega^2}$$
$$|c_3(t)|^2 = \frac{e^{-t\Gamma}\Gamma^2(\cosh(\Omega t) - 1)}{2\Omega^2}$$

$$\Gamma < \omega, \qquad |c_2(t)|^2 = \frac{e^{-t\Gamma}((1+\cos(\Omega t))\Gamma^2 - 2\omega^2)}{2\Omega^2}$$
$$|c_3(t)|^2 = \frac{e^{-t\Gamma}\Gamma^2(\cos(\Omega t) - 1)}{2\Omega^2}$$

with $\Omega = \sqrt{|\Gamma^2 - \omega^2|}$.
For $\Gamma = 0$, we just get $\tilde{c}_2(t) = e^{-i\omega_{23}t}$ and $\tilde{c}_3(t) = 0$, which is just the free evolution of the energy states as $|\tilde{2}\rangle$ is ω_{23} above $|\tilde{3}\rangle$ in energy. In this case, the system stays in the atomic states and no photons are emitted. We see that the populations have an expontial decay envelope with decay rate Γ. Thus, Γ indicates how often the excited states spontaneously emit a photon.

Exercise 1d

With $\omega_{23} = 0$, Equation 1 becomes

$$\dot{\tilde{c}}_2 = -\Gamma/2(\tilde{c}_2 + \tilde{c}_3)$$
$$\dot{\tilde{c}}_3 = -\Gamma/2(\tilde{c}_2 + \tilde{c}_3)$$

which yields

$$\frac{d}{dt}(\tilde{c}_2+\tilde{c}_3) = -\Gamma/2(\tilde{c}_2+\tilde{c}_3) \implies \tilde{c}_2+\tilde{c}_3 = (\alpha/2)e^{-\Gamma t/2}$$
$$\frac{d}{dt}(\tilde{c}_2-\tilde{c}_3) = 0 \implies \tilde{c}_2-\tilde{c}_3 = \beta/2$$

Where α and β are determined by initial conditions. With this, we find

$$\tilde{c}_2(t) = \alpha e^{-\Gamma t/2} + \beta$$
$$\tilde{c}_3(t) = \alpha e^{-\Gamma t/2} - \beta$$

For complete decay, we need $\beta = 2(\tilde{c}_2(0) - \tilde{c}_3(0)) = 0 \implies \tilde{c}_2(0) = \tilde{c}_3(0)$. If we have no population in the $|k\rangle$ states to begin, then normalization requires $1 = |\tilde{c}_2(0)|^2 + |\tilde{c}_3(0)|^2 = 2|\tilde{c}_2(0)|^2$ so that the superposition which decays completely is given by

$$|\psi_{\text{decay}}\rangle = \frac{|2\rangle + |3\rangle}{\sqrt{2}}$$

unique up to a global phase. For no decay at all, we need $\alpha = 0$. Similar reasoning leads us to an antisymmetric state which does not decay.

$$|\psi_{\text{no decay}}\rangle = \frac{|2\rangle - |3\rangle}{\sqrt{2}}$$

This state corresponds to when a photon decays from one atomic state and then reabsorbed to bring it to the other state.
Beat frequency must depend on Γ and ω_{23}. Since Γ controls decay rate, it is likely not involved. All that's left is ω_{23} so would assume beat frequency is $\propto \omega_{23}$.

Exercise 2a

Let $\chi(t) = \int_0^t f(t')dt'$. For $s(t) = 1$, the Hamiltonian looks like

$$\hat{H}(t) = \hbar\omega\hat{S}_z + \hbar f(t)\hat{S}_z$$

With this, the expectation of $\hat{S}_+ = |2\rangle\langle 1|$ is given by

$$\langle \hat{S}_+ \rangle = e^{i\omega t}\langle e^{i\chi(t)}\rangle \rho_{12}(0)$$

Since χ is an integral of gaussians, it is itself gaussian, with 0 mean and varianec $\sigma_t^2 = \langle \chi(t)^2 \rangle$. Thus,

$$\langle e^{i\chi(t)}\rangle = \int_{-\infty}^{\infty} \frac{e^{-\chi^2/2\sigma_t^2}}{\sqrt{2\pi\sigma_t^2}} e^{i\chi} d\chi = \mathcal{F}\left\{ \frac{e^{-\chi^2/2\sigma_t^2}}{\sqrt{2\pi\sigma_t^2}} \right\}(t=1) = e^{-\sigma_t^2/2}$$

Thus, it remains to calculate σ_t^2

$$\sigma_t^2 = \langle \int_0^t f(t')dt' \int_0^t f(t'')dt'' \rangle = \int_0^t dt' \int_0^t \langle f(t')f(t'')\rangle dt''$$

Changing variables to $\tau = t' - t'', d\tau = -dt''$ in the inner integral, and using the assumption $\langle f(t+\tau)f(t)\rangle = \langle f(\tau)f(0)\rangle = \Delta^2 e^{-|\tau|/\tau_c}$, we find

$$\sigma_t^2 = -\Delta^2 \int_0^t dt' \int_{t'}^{t'-t} e^{-|\tau|/\tau_c} d\tau = 2\Delta^2\tau_c(t - \tau_c(1 - e^{-t/\tau_c}))$$

where the integral was evaluated using mathematica (see accompanying notebook). We examine the limits

(i) Short Correlation Time ($\tau_c \ll t$): In this limit $e^{-t/\tau_c} \to 0$ so $\sigma_t^2 \to 2\Delta^2\tau_c(t - \tau_c)$ and we get

$$\langle \hat{S}_+ \rangle_{\tau_c \ll t} = e^{i\omega t} e^{-\Delta^2\tau_c(t-\tau_c)} \rho_{12}(0)$$

(ii) Long Correlation Time ($\tau_c \gg t$): Here, $e^{-t/\tau_c} \approx 1$ so that $\sigma_t^2 \to 2\Delta^2\tau_c t$. Hence,

$$\langle \hat{S}_+ \rangle_{\tau_c \gg t} = e^{i\omega t} e^{-\Delta^2\tau_c t} \rho_{12}(0)$$

We see that $\langle \hat{S}_+ \rangle_{\tau_c \ll t} = e^{\Delta^2\tau_c^2} \cdot \langle \hat{S}_+ \rangle_{\tau_c \gg t}$ and in both cases we can identify $T_2^* = 1/\Delta^2\tau_c$. In the limit $\tau_c \to 0$, we find $\sigma_t^2 \to 0$ and $T_2^* \to \infty$, implying that the system does not lose coherence? The noise looks white in this limit, and it seems that the environment fluctuations end up canceling each other?

Exercise 2b

The propagator for the first half of the experiment, where $s(t) = +1$, looks like

$$\begin{aligned}\hat{U}^+ &= \exp(-\int_0^{t/2} \frac{i}{\hbar} H(t')dt') \\ &= \exp(-i\int_0^{t/2} (\omega + f(t'))\hat{S}_z dt') \\ &= \exp(-i(\omega t/2 + \int_0^{t/2} f(t')dt')\hat{S}_z)\end{aligned}$$

Whereas, the propagator for the second half, where $s(t) = -1$ causing $H(t)$ to flip sign, looks like

$$\hat{U}^- = \exp(i(\omega t/2 + \int_{t/2}^{t} f(t')dt')\hat{S}_z)$$

The total propagator, $\hat{U}(t)$, is then

$$\hat{U}(t) = \hat{U}^-\hat{U}^+ = \exp(i(\int_{t/2}^{t} f(t')dt' - \int_0^{t/2} f(t')dt')\hat{S}_z)$$

Then, we find

$$\langle S_+ \rangle = \langle e^{i(\int_{t/2}^{t} f(t')dt' - \int_0^{t/2} f(t')dt')} \rangle \cdot S_+(0)$$

Letting $\chi(t) = \int_{t/2}^{t} f(t')dt' - \int_0^{t/2} f(t')dt'$ we have as before that $\langle e^{i\chi(t)} \rangle = e^{-\sigma_t^2/2}$. With σ_t^2 calculated as

$$\begin{aligned}\sigma_t^2 &= \langle \chi(t)^2 \rangle \\ &= \langle (\int_{t/2}^{t} f(t')dt' - \int_0^{t/2} f(t')dt')(\int_{t/2}^{t} f(t'')dt'' - \int_0^{t/2} f(t'')dt'') \rangle\end{aligned}$$

This computation reduces to two products

$$\begin{aligned}\sigma_1^2(t) &= \int_0^{t/2} dt' \int_0^{t/2} \langle f(t')f(t'') \rangle dt'' \\ \sigma_2^2(t) &= \int_0^{t/2} dt' \int_{t/2}^{t} \langle f(t')f(t'') \rangle dt''\end{aligned}$$

Then, $\sigma_t^2 = 2\sigma_1^2 - 2\sigma_2^2$. Using mathematica (see accompnying notebook), we find

$$\sigma_t^2 = 2\Delta^2\tau_c(t - (3 + e^{-t/\tau_c} - 4e^{-t/2\tau_c})\tau_c)$$

In $\tau_c \ll t$, we get $\sigma_t^2 \to 2\Delta^2\tau_c(t - 3\tau_c)$ so the decay rate is still $1/\Delta^2\tau_c$. For $\tau_c \gg t$, get same result?? Maybe expand to higher orders, or something wrong.
Long correlation time means the environment is not fluctuating much during the timescale of the experiment. In this case, the spin echo is reliable because the effect of the environment can be undone by the π pulse. For shorter correlation times, the fluctuations are too large to undo.

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